MSc DISSERTATION · UNIVERSITY OF EDINBURGH · EPCC 2025–26

Hierarchical Deep Learning for Fake News Detection

A taxonomy-driven BERT architecture for fine-grained misinformation classification on the Fakeddit dataset

564K
Training Samples

84%
Accuracy

0.81
Macro F1

8
Categories

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MOTIVATION

Why Does This Matter?

Scale of the Problem

Fakeddit contains 1M+ Reddit posts spanning 6 misinformation categories — far beyond human review capacity.

Existing Approaches Fall Short

Most classifiers treat fake news as binary (real/fake). Real misinformation spans distinct strategies that require fine-grained distinction.

Hierarchical Structure is Underexplored

Categories naturally cluster into semantic groups. Exploiting this hierarchy should reduce cross-group confusion and improve minority class performance.

DATASET & TAXONOMY

Fakeddit + Extended Taxonomy

Fakeddit Dataset

564,000	Training samples
59,319	Test samples
6	Original categories
2020	Reddit data vintage

Group 0 — Authentic

True Content

Group 1 — Structural Deception

Misleading · False Connection · Clickbait*

Group 2 — Fabricated/Manipulated

Satire · Imposter · Manipulated · AI-Generated*

** Categories added as part of this dissertation's taxonomy extension*

ARCHITECTURE

Two-Stage Hierarchical BERT

BERT Encoder

bert-base-uncased
768d CLS token

Coarse Head

3-way classifier
→ Group 0/1/2

Why: Shared representations reduce cross-group confusion

G0 Fine Head

1-class
True Content

G1 Fine Head

3-class
Misleading/FalseConn/Clickbait

G2 Fine Head

4-class
Satire/Imposter/Manip/AI

Joint Loss = 0.5 × Coarse Loss + 0.5 × Fine Loss

Why joint: end-to-end training ensures coarse and fine heads improve together

EXPERIMENT 1

Hierarchical vs Flat Baseline

Same BERT backbone · Same data · Same hyperparameters · 3 seeds · p < 0.05

Freezing Strategy

BERT frozen at lr=1e-5 · Heads at lr=2e-5 · Prevents overwriting pretrained representations

Flat Baseline (Macro F1: 0.75)

Hierarchical (Macro F1: 0.77) · p < 0.05

Key Finding

Outperforms flat on 5/6 categories

Satire +0.04
Imposter +0.04
Manipulated +0.03

→ *Hierarchy helps most for minority classes*

EXPERIMENT 2

Taxonomy Extension: AI-Generated Content

Attempt 1

artem9k/ai-text-detection-pile

F1: 0.11

Recall: 0.18

Long essays — severe domain mismatch vs Fakeddit short Reddit titles

Why: Domain mismatch: model trained on 500+ word essays fails on 8-word headlines

Attempt 2

artnitolog/llm-generated-texts

F1: 0.71

Recall: 0.61

Reuters headlines — better length but still some mismatch

Why: Shorter inputs closer to Fakeddit style — confirms domain mismatch hypothesis

Attempt 3

gsingh1-py/train (NYT titles)

F1: 0.90

Recall: 0.86

Short NYT headlines — matched Fakeddit title length precisely

Why: Short headline format matches training distribution — optimal domain alignment

CURRENT STATUS & NEXT STEPS

Analysis Complete · Clickbait Extension In Progress

Error Analysis Findings

74% of all errors are coarse routing errors — the coarse head is the bottleneck

55% error rate on Imposter Content — most challenging category

0.924 model confidence when wrong on Manipulated Content — overconfident

Next: Clickbait Extension

01 New Category in Group 1

Clickbait placed under Structural Deception — uses misleading framing, not fabrication. Label 7 added to taxonomy.

02 Dataset: marksverdhei/clickbait

16K clickbait headlines from real news. Augmented 10× to balance against majority classes.

03 Selective Freezing Strategy

BERT + Group 2 head frozen. Only Group 1 head + coarse head retrained. Preserves existing performance.

04 Expected Outcome

Target F1 > 0.80 on Clickbait. Monitor for regression on Misleading + False Connection.

CHALLENGES & RISKS

Technical Challenges & Fallbacks

Class Imbalance

Imposter Content = 2.1% of data. Risk of model ignoring minority classes.

✓ Weighted cross-entropy loss (Imposter 5×, Manipulated 4×, Satire 3×)

Catastrophic Forgetting

Full retraining risks overwriting existing category knowledge.

✓ Selective freezing — only retrain heads relevant to new category

Domain Shift

External datasets (AI text, Clickbait) have different style from Reddit posts.

✓ Iterative dataset refinement — proven in AI-Generated extension (F1: 0.11→0.90)

Compute Constraints

Full 564K dataset training takes 6+ hours on T4 GPU.

✓ 200K subsample for extensions; full training for final model

SUCCESS CRITERIA

What Does Success Look Like?

COMPLETE

Hierarchical outperforms flat baseline

Achieved: Macro F1 0.77 vs 0.75 ($p < 0.05$, 3 seeds)

COMPLETE

AI-Generated taxonomy extension F1 > 0.85

Achieved: F1 0.90 after 3 iterative dataset refinements

IN PROGRESS

Clickbait extension F1 > 0.80

Implementation underway — dataset sourced, freezing strategy defined

PLANNED

Multimodal extension — text + image

ResNet-50 + BERT late fusion. Requires EIDF GPU allocation.

PLANNED

Full error analysis & model improvement

Systematic misclassification study + weighted loss retraining

PLANNED

Distinguishability

Practical ceiling exists. AI-Generated vs Manipulated overlap is the hardest pair. Text-only models need multimodal signals to go finer.

SUMMARY

🏗 Architecture

Two-stage hierarchical BERT — coarse group head + group-specific fine heads. Joint loss enables end-to-end learning.

📊 Results to Date

84% accuracy · Macro F1 0.81 · Hierarchical beats flat on 5/6 categories · AI-Generated F1 0.90 after iterative refinement.

🔧 In Progress

Clickbait as 8th category · Error analysis-driven model improvement · Weighted loss for class imbalance.

🚀 Planned + Live Demo

Multimodal extension (text + image) · Out-of-distribution evaluation · Live demo live at huggingface.co/spaces/imaniuoboubouv/fake-news-detector